

Projecting Pitcher Strikeout Percentage

Executive Summary — 2025 Projections

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The problem is regression, not regressors

K% is a binomial rate observed over wildly unequal samples. The panel holds 4,371 player-seasons for 1,743 pitchers. Batters faced run from 1 to 886, and 33% of player-seasons fall below the roughly 70 batters faced at which K% stabilises. A reliever with 40 batters faced and a starter with 800 both report a K%, but one is mostly noise and the other is mostly signal. The decision that drives accuracy is therefore how much of each pitcher's observed rate to believe, not which regressor to add. I treat that as the modelling problem and derive the answer from the data rather than assuming it.

The model

An empirical-Bayes projection in the Marcel and Steamer tradition, fitted on 2021-2024 only. Decomposing the variance of observed K% into true talent and binomial noise gives the regression constant directly:

$$\text{Var}(\text{observed}) = 0.00298 = \text{Var}(\text{talent}) 0.00228 + \text{binomial noise } 0.00070 \rightarrow k = 77 \text{ batters faced}$$

Each pitcher's record is shrunk toward a prior worth 77 batters faced. That constant is measured, not chosen, and it independently reproduces the published stabilisation point for K% of roughly 70 batters faced — two different routes to the same number, which is the main evidence that the shrinkage is right. Five components sit on top of it:

- **A Stuff+-informed prior.** Stuff+ becomes reliable in far fewer pitches than K% does, so the prior is $E[K\% \mid \text{Stuff+}, \text{age}]$ rather than a flat league average. It must be calibrated forward first: the contemporaneous prior is over-dispersed when applied to the next season (slope 0.75), and uncorrected it performs worse than a flat prior. A more informative prior also earns a larger effective sample size, 136 batters faced against 77.
- **Fitted recency weights.** Marcel's 5/4/3 is a starting grid, not an assumption; a search over the training seasons selects 1.00/0.60/0.35.
- **A league-detrended aging curve,** estimated by the delta method on paired seasons and applied as a one-year nudge rather than extrapolated, because it is survivorship-biased by construction.
- **League re-anchoring,** moving each pitcher's record from the environment it was set in into the projected 2025 environment of 22.58%.
- **A 50/50 blend with a ridge model** on engineered features, chosen a priori on the reasoning behind FanGraphs Depth Charts and ATC, not because it won the backtest.

Held-out result: 2025 was never used to fit anything

Model	RMSE	MAE	R2
League average	5.29 pp	4.15 pp	-0.000
Last season's K%, unregressed	5.72 pp	4.30 pp	-0.169
Empirical Bayes, flat prior	4.10 pp	3.18 pp	0.401
Empirical Bayes, Stuff+ prior	4.08 pp	3.16 pp	0.407
Ridge on engineered features	4.06 pp	3.22 pp	0.410
Blend of EB + ridge (shipped)	4.04 pp	3.15 pp	0.418

Pitchers with at least 100 batters faced in 2025 ($n = 459$). The shipped model cuts RMSE 24% against a league-average baseline and 29% against last season's raw rate, which is worse than the league average because unregressed rates carry their full sampling noise forward.

What each component is worth

Variant	RMSE	R2	Cost (bps)
FULL empirical-Bayes model	4.08 pp	0.407	--
no aging adjustment	4.15 pp	0.384	+7.7
no league re-anchoring	4.07 pp	0.407	-0.1
no recency weighting (1/1/1)	4.19 pp	0.372	+11.8
flat prior (Stuff+ unused)	4.10 pp	0.401	+2.1
uncalibrated Stuff+ prior	4.13 pp	0.390	+5.8
almost no shrinkage (k=5)	4.90 pp	0.143	+82.3
over-shrunk (k=400)	4.10 pp	0.399	+2.6

Each component disabled in turn, re-scored on held-out 2025. Shrinkage is the load-bearing decision by an order of magnitude: setting $k = 5$ costs 82 basis points, against 12 for dropping recency weighting and 8 for dropping aging. Over-shrinking is far cheaper than under-shrinking, which is the asymmetry the whole design leans on.

Validation

Leakage guard. The instructions bar data from Opening Day 2025 onward, but the supplied file contains a complete 2025 season. Those rows are treated strictly as an answer key: read once, after the projections are written, to score them. Every fitted object — league curve, shrinkage constant, aging curve, Stuff+ prior and its calibration, ridge, LightGBM and the recency weights — is estimated on 2021-2024 alone. `--self-test` re-runs the entire pipeline against a copy of the file with every 2025 row deleted and asserts the projections are unchanged. They are, to 0.00e+00 across all 1,328 projections and every numeric column.

Rolling-origin backtest. For each target season 2023-2024 the pipeline is re-fitted on transitions completed before it, including a fresh recency-weight search inside each fold, so the table is out-of-sample with respect to the weight search as well as the outcome. Model ranking is stable across folds and thresholds; fold-to-fold separation among the top models is inside noise, which is precisely why the shipped model is a blend rather than the backtest winner.

Calibrated uncertainty. Every projection ships with an 80% prediction interval combining talent uncertainty, which shrinks as reliability rises, with the binomial uncertainty of next season's observed rate. The second term dominates for low-usage pitchers. Empirical coverage on held-out 2025 is 81.5% against a nominal 80%. Interval widths are calibrated using projected playing time, never the realised batters faced, which is unknowable when the interval is issued.

Honest limitations

- The aging curve is survivorship-biased: pitchers who decline are not re-observed, so the curve understates true decline. It is applied as a single one-year step and never extrapolated across a career.
- Starters and relievers are pooled. Role changes move K% by several points, and a role-aware prior is the single most promising extension the supplied columns would support.
- Playing time is projected crudely and used only to size intervals, never as a forecast. A real depth-chart input would sharpen the low-usage tail.
- Four seasons of history support at most two honest backtest folds. The held-out result rests on one season, and the gaps among the top three models (4.10, 4.08 and 4.04 pp) are small relative to that.
- Model selection was performed by a human who has seen the held-out season. The self-test proves no 2025 row reaches a fitted object; it cannot prove the same of the analyst's judgement. The blend was fixed a priori for this reason.

Deliverables

`k_pct_projections_2025.csv` carries 1,328 projections with 80% intervals, reliability and the component decomposition behind each one. `methodology_2025.pdf` is the full derivation. `backtest_2025.csv`, `ablation_2025.csv` and `run_manifest_2025.json` record every validation number and every fitted constant for auditing. The analysis is deterministic: fixed seed, no network, no data beyond the supplied panel.